

Suryansh Singh Rawat

☎ (206) 496 5588 ✉ suryansh@uw.edu 🌐 [suryansh-singh-rawat](https://suryansh-singh-rawat.github.io) 🐙 [xsuryanshx](https://xsuryanshx.github.io) ✉ [snareyansh](https://snareyansh.github.io) 🌐 suryansh.space

OBJECTIVE

AI/ML Engineer with 4 years of experience building enterprise-scale RAG, agentic LLM systems and cybersecurity NLP research, focused on shipping production-grade AI systems at scale. Currently Seeking Summer 2026 MLE / AI Research Intern roles.

EDUCATION

University of Washington, Seattle

Sept 2025 – Mar 2027

Master of Science (M.S), Data Science | **GPA: 3.81/4.0**

Seattle, WA

- Relevant Courses: Natural Language Processing, Machine Learning, Probability and Statistics, Applied Statistics and Experimental Design, Software Design for Data Science, Scalable Data Systems and Algorithms

Birla Institute of Technology and Science, Pilani

Aug 2018 – July 2022

Bachelor of Engineering (B.E), Electronics and Instrumentation

Goa, India

TECHNICAL SKILLS

Languages: Python, C++, SQL, R, Java, TypeScript, Bash, MATLAB

Developer Tools: Git, AWS, Microsoft Azure, Docker, S3, DynamoDB, PostgreSQL, MongoDB, Redis, Supabase

Frameworks: PyTorch, TensorFlow, Keras, SpaCy, HuggingFace, LangChain, LangGraph, CrewAI, Streamlit, FastAPI, Selenium

WORK EXPERIENCE

Ascentt

Mar 2024 – Sept 2025

Principal AI Engineer

Remote, US

- Scaled ToyotaGPT (ChatGPT equivalent for enterprises) to **74,000+** active users, architecting **REST APIs**, **DynamoDB** for user state management, and an **Agentic RAG** pipeline using **LangGraph** managing **1TB+ of unstructured data** across **14+** document types.
- Optimized RAG pipelines across **4 enterprise** products, implementing lossless preprocessing, chunking, and indexing in **Amazon OpenSearch**, and benchmarking with **RAGAS**, achieving **+25% Precision@10** and **+12% Recall@10** over vanilla RAG baseline.
- **Led backend repository maintenance** and code reviews, ensuring high-quality, production-ready code across enterprise systems.

Deloitte USI

July 2022 – Mar 2024

Data Scientist | AI Center of Excellence Team

Bangalore, India

- Developed **Autoencoder anomaly detection** models with graph-based preprocessing in **Neo4J** to detect **zero-day threat attacks** from real-time AWS network flow data.
- Researched an NLP pipeline for regulatory document consolidation, leveraging **metric learning** trained **SecBERT** embeddings and **HDBSCAN** to merge semantically overlapping compliance requirements, achieving **~83% reduction** in manual SME efforts.
- Constructed a **SecBERT text summarization model** for generating cyber threat reports, reducing analyst review time by **~40%**, integrating **NER tagging and masking** with **SpaCy** annotation, deploying end-to-end pipeline using **Flask and Docker**.

Scienaptic AI

July 2021 – Dec 2021

Data Scientist Intern

Bangalore, India

- Achieved an **11.5% increase** in application approvals by building credit underwriting models using **XGBoost and Logistic Regression** to forecast the probability of credit accounts that may default.

SELECTED PROJECTS

OpenProbe 🐍 | Python, Langchain, LangGraph

May 2025 – June 2025

- Built OpenProbe, a deep research **multi agent system** using **LangGraph** for state orchestration (websearch, code, reasoning).
- Achieved **67.1% accuracy** on the FRAMES benchmark, outperforming GPT-4o-Search by **1.5%**, using DeepSeek-R1/Qwen3-32B.

Ad-Astra: xAI Ad Intelligence Engine 🐍 | Python, Grok, XAPI, Supabase

Dec 2025 – Dec 2025

- Built a personalized ad generation and placement engine for X (Twitter), using **Grok 4.1 Reasoning & Vision models** for persona-aware ad copy generation and **scroll telemetry** to optimize in-feed placement; improving engagement and ad relevance.

PersonaAI 🐍 | Python, Microsoft Phi-4, Supabase

Oct 2025 – Oct 2025

- Built an **AI simulation platform** to test digital campaigns before launch, running **A/B tests** using **NVIDIA Nemotron** Personas.
- Created natural-language audience targeting with diagnostics explaining why variants succeed or fail across segments.

KeyCognition 🐍 | Python, PyTorch, React

Nov 2025 – Nov 2025

- Built a behavioural analysis platform processing **136M+ keystroke events** to predict user cognitive load (low/med/high).
- Leveraged temporal typing features (e.g. using backspace) with **LSTM and K-Means models** for user behavior classification.

ACHIEVEMENTS

- Placed **3rd out of 40+ teams** at the **Lovable.dev Seattle Hackathon 2025** for building **PersonaAI**.
- Placed **3rd out of 20+ teams** at the **UW Databricks Hackathon 2025** for building **KeyCognition**.
- Recipient of **\$5000 Thinking Machines Research Grant** for proposed research on improving LLM Planning and Personalization.